



MINING DEALER BEST PRACTICE

“Plug and Play” System for Engine Readiness on the Dynamometer

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

A good readiness process of the engine on the dynamometer at the time of the performance tests is key to ensuring the correct assessment of all operational parameters.

The large number of electrical, hydraulic and pneumatic connections and the great lengths of cables and hoses increase the risk of accidents for the technicians and for the components as well. They also originate connection errors and increase the engine readiness time. The aforementioned gives a disorganized appearance during the engine setting before running the test. See Fig. 1.

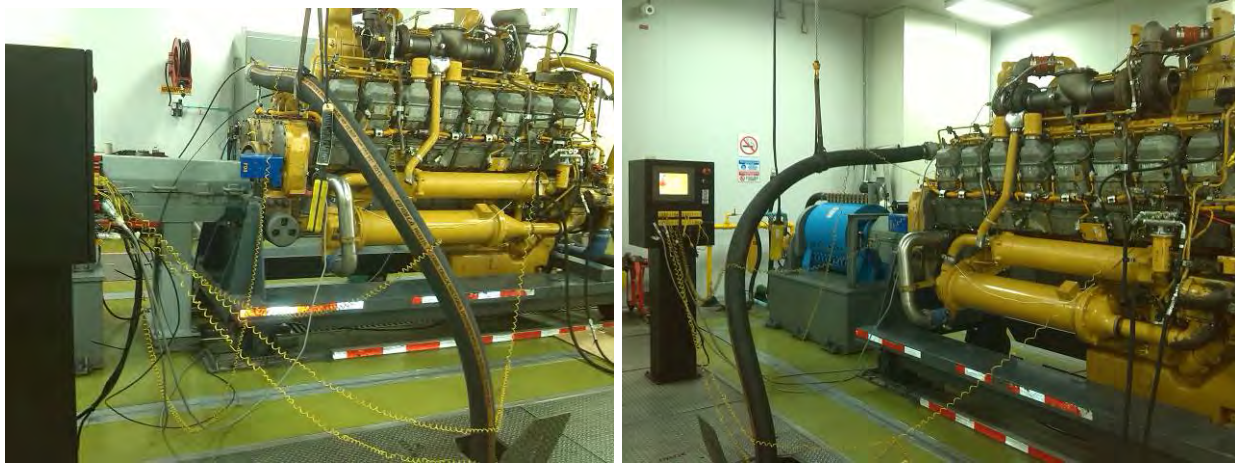


Fig. 1. Engine electrical connection before the upgrade.

This Best Practice aims to improve the aforementioned problems thru the design of a plug and play system for the various connections required at the engine readiness stage when testing on the dynamometer.

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2.0 Best Practice Description

This improvement consists of quick, safe and orderly “plug and play” connections, for the installation of the hydraulic, pneumatic and electrical lines that the engine requires when it is tested during its readiness stage on the dynamometer. This routing of lines from the SISWEB workstation to the dynamometer is done through internal channel drains of the room; centralizing all the terminals in a single panel on one side and very near the equipment and the engine so as to reduce the lengths and organize the final connecting bridges to the engine. See Fig. 2.

The panel basically consists of the relocation of the following connections listed in the following chart:

PLUG AND PLAY CONECTORS		
1	P1	ENGINE OIL PRESSURE INLET
	P2	ENGINE FUEL PRESSURE INLET
	P3	ENGINE BOOST PRESSURE INLET
	P4	TEST POINT PRESSURE INLET
2	T	ENGINE TEMPERATURES INLET
3	ECM	ELECTRONIC CONTROL MODULE CONNECTOR
4	HYDAC IN	HIGH PRESSURE FROM ENGINE INLET (PARTICLE COUNTDOWN)
	HYDAC OUT	LOW PRESSURE TO ENGINE OUTLET(PARTICLE COUNTDOWN)

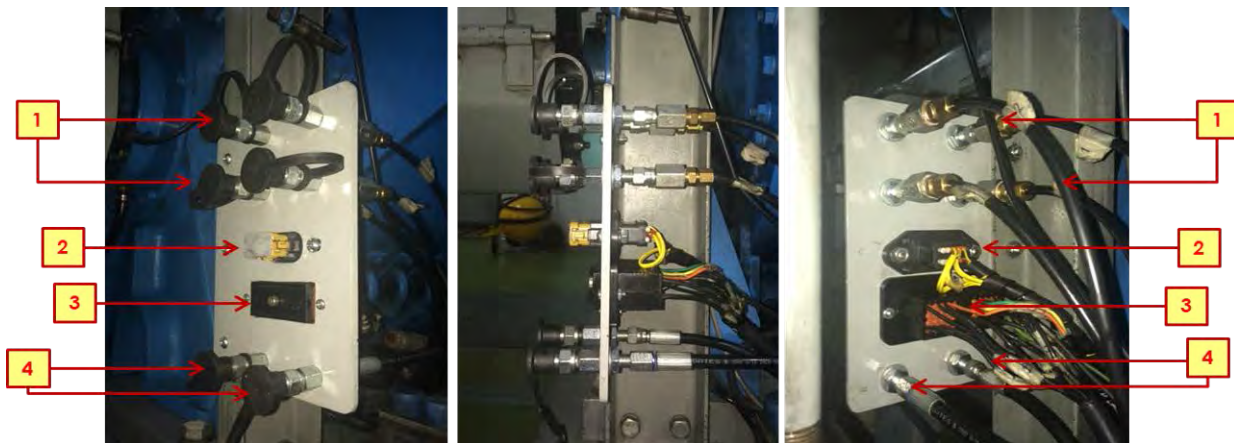


Figure 2.

3.0 Implementation Steps

From the workstation ports (A) the hoses to the pressure taps of the lubrication system, fuel and air are connected; and the cables extension for the thermocouples, are taken to the channel drains, enter the point (B) and exit through the point (C), then by a metal conduits are routed up to the plate (D), which is anchored by bolts to a channel welded to the dynamometer base. See Fig. 3.

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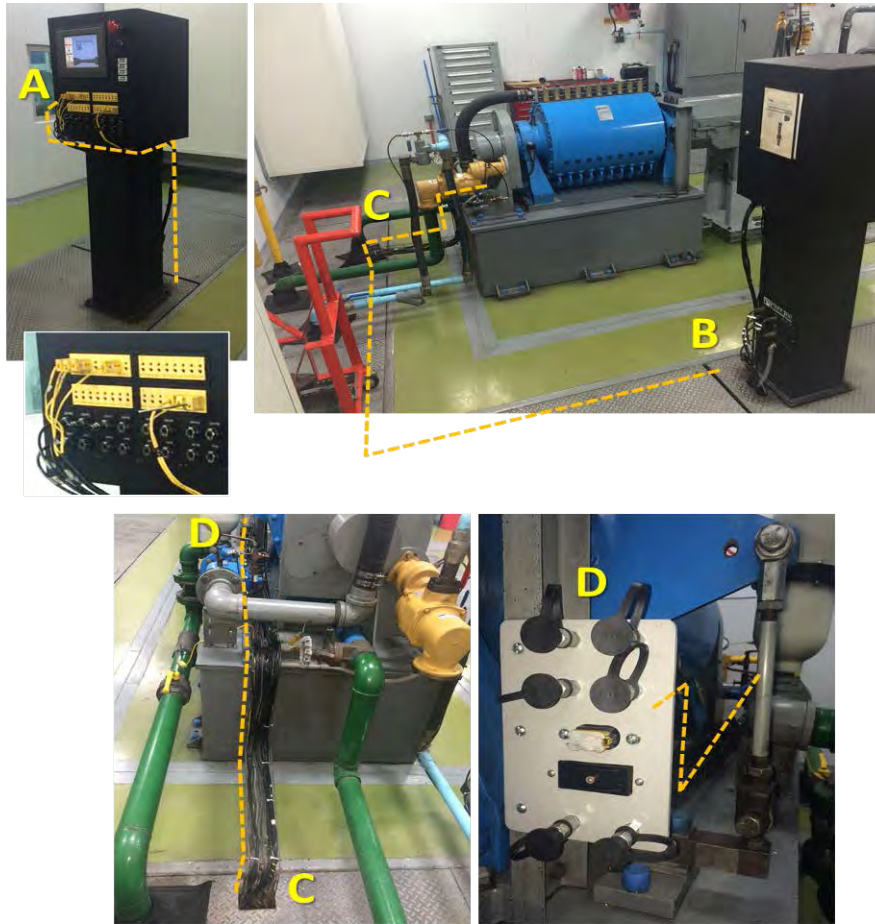


Figure 3.

4.0 Benefits

The benefits that come with this best practice are safety oriented, as it seeks to eliminate the risk of falling by tripping due to the large number of electrical, hydraulic and pneumatic connections, which are organized and protected. Thanks to the implementation of plug and play the risk of making wrong connections are greatly diminished, leading to a faster readiness process when running the test. 3.

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5.0 Resources Required

List of specific accessories for each connection. See Fig. 4:

ID	P/N	Description	QTY
1	N/A	Plate (6.3 * 170 *240) mm	1
2,5	8T-3613	Nipple AS valved	6
6	4U-1697	Adaptor bulkhead	6
6A	5B-3254	Nut	6
7	145-8833	Union Adapter	6
-	6V-4144	Coupler AS	6
11	7X-7481	Thermocouple Wire - Bulk (m)	4
10	103-8134	Receptacle (12 pins)	1
3	155-2255	Plug kit connector (12 pins)	1
12	9X-1343	Receptacle (40 pins)	1
4	8T-9834	Connector Plug (40 pins)	1
-	126-1767	Pin - Connector	12
-	126-1768	Socket - Connector	12
9	N/A	Hose - .25" OD	40m
8	N/A	Brass Fitting - .25" * .25"	4

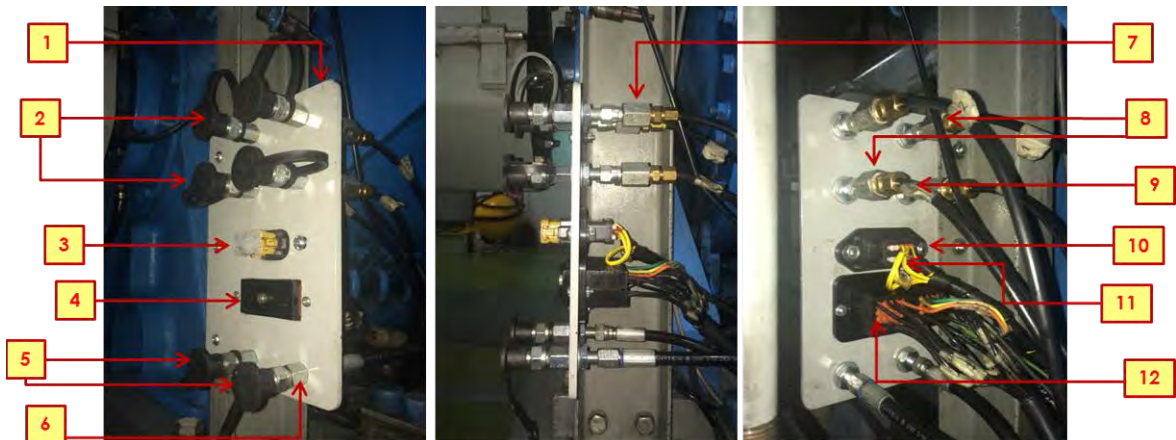


Figure 4.

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6.0 Supporting Attachments / References

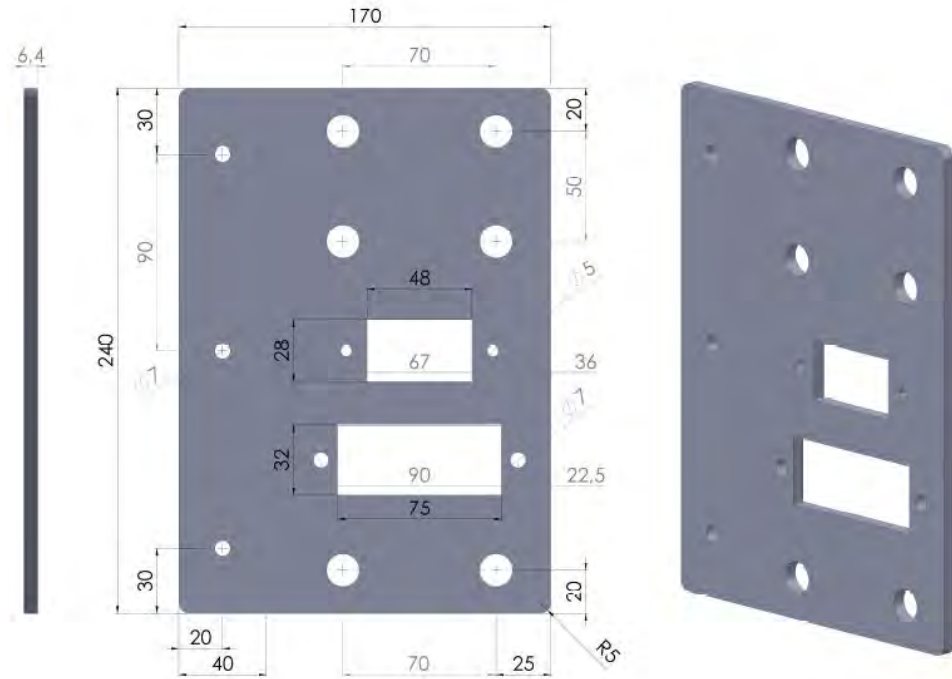


Figure 5.

7.0 Related Best Practices

- 0906-4.4-1014 - Dyno Room Fluid Lines

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8.0 Acknowledgements

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